SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE CODE: ITA 06 **COURSE NAME: MACHINE LEARNING**

COURSE OUTCOME COVERED

CO2: *Neural Network Representation, Perceptron Model, Multilayer Perceptron's, Backpropagation Algorithm, Deep Neural Networks, Genetic Algorithms, Hypothesis Space Search, Genetic Programming, Models of Evaluation and Learning (BL1, BL2)*

Assessment Tool Weightage: 10%

QUESTIONS:15 **Total Marks:100**

Annexure A

DEBUGGING & OPTIMIZATION TASK QUESTIONS

Code of Conduct:

I ______Bandaru Jayasri-192424301___ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Criteria	*Conceptual Understanding*	*Accuracy of Answers*	*Application of Knowledge*	*Completeness of Response*	*Clarity & Presentation*	*Total*
Weightage	*30%*	*20%*	*30%*	*10%*	*10%*	
Q1						
Q2						
Q3						

Signature of the student: ***Total Marks Awarded***
B.Jayasri

Annexure A

Short Answer Test

1. A perceptron model is trained on a dataset that is known to be linearly separable, yet the algorithm fails to converge even after many iterations. Analyze the possible causes of this issue, including improper learning rate selection, incorrect weight initialization, bias handling errors, or flawed implementation of the update rule. Propose debugging strategies to identify the root cause and suggest optimization techniques to ensure faster and stable convergence. (5 Marks)

2. While training a multilayer perceptron using the backpropagation algorithm, the network exhibits slow convergence and gets stuck in local minima. Examine the potential reasons such as vanishing gradients, inappropriate activation functions, poor weight initialization, or suboptimal learning rates. Describe step-by-step debugging approaches to trace the issue and recommend optimization techniques like adaptive learning rates, normalization, or advanced optimizers to improve performance.
(5 Marks)

3. A genetic algorithm designed for hypothesis space search is producing inconsistent and suboptimal solutions across multiple runs. Investigate possible issues such as improper encoding of chromosomes, incorrect fitness function design, premature convergence, or lack of diversity in the population. Explain how to debug these problems systematically and suggest optimization strategies like mutation tuning, selection pressure adjustment, and elitism to enhance solution quality and convergence speed. (5 Marks)

4. A deep neural network combined with genetic programming is used for solving a complex prediction problem, but it suffers from overfitting and high computational cost. Identify debugging steps to analyze issues related to model complexity, over-parameterization, and insufficient training data. Propose optimization techniques such as regularization, dropout, pruning, and efficient evaluation models to balance accuracy and computational efficiency while improving generalization capability.
(5 Marks)

5. A decision tree model trained on a large dataset shows high accuracy on training data but poor performance on unseen data. Analyze possible causes such as overfitting, deep tree structure, or noisy data. Propose debugging steps and optimization methods like pruning, cross-validation, and feature selection to improve generalization. (5 Marks)

6. A convolutional neural network (CNN) used for image classification produces unstable results across training epochs. Examine potential issues such as improper data augmentation, vanishing gradients, or batch normalization errors. Describe debugging strategies and suggest optimization techniques to stabilize training and improve accuracy. (5 Marks)

7. A reinforcement learning agent fails to learn an optimal policy in a simulated environment. Investigate causes such as poor reward design, insufficient exploration, or incorrect state representation. Explain debugging methods and recommend improvements like reward shaping, exploration strategies, and tuning hyperparameters. (5 Marks)

8. A clustering algorithm like K-means produces inconsistent cluster assignments for similar datasets. Analyze possible reasons such as poor initialization, incorrect number of clusters, or sensitivity to outliers. Propose debugging approaches and optimization strategies like K-means++, normalization, and silhouette analysis.
(5 Marks)

9. A support vector machine (SVM) model underperforms on a nonlinear dataset. Identify issues such as incorrect kernel choice, improper parameter tuning, or scaling problems. Suggest debugging techniques and optimization methods including kernel selection, hyperparameter tuning, and feature scaling. (5 Marks)

10. A natural language processing (NLP) model using recurrent neural networks (RNNs) suffers from poor long-term dependency learning. Examine causes such as vanishing gradients and limited memory capacity. Describe debugging steps and propose solutions like LSTM/GRU units, attention mechanisms, and improved training strategies. (5 Marks)

11. A support vector machine (SVM) model trained for classification is producing very low accuracy on both training and test datasets. Analyze possible causes such as improper kernel selection, incorrect hyperparameter tuning (C and gamma), or poor feature scaling. Describe systematic debugging steps to identify the issue and suggest optimization techniques like kernel tuning, normalization, and grid search to improve model performance. (5 Marks)

12. A K-Means clustering algorithm is applied to a dataset, but the clusters formed are highly imbalanced and do not reflect the actual data distribution. Investigate possible issues such as incorrect choice of K, poor initialization of centroids, or presence of outliers. Explain debugging approaches and suggest improvements like the Elbow method, K-Means++, and outlier removal techniques. (5 Marks)

13. A recurrent neural network (RNN) designed for time-series prediction shows inconsistent outputs and fails to capture long-term dependencies. Analyze possible causes such as vanishing gradients, insufficient sequence length, or improper architecture design. Describe debugging steps and propose optimization techniques such as using LSTM/GRU units, gradient clipping, and sequence normalization. (5 Marks)

14. A Naïve Bayes classifier used for text classification performs poorly when applied to real-world data containing noisy and irrelevant features. Examine possible issues such as violation of feature independence assumption, improper preprocessing, or imbalanced dataset. Explain debugging strategies and suggest improvements like feature selection, smoothing techniques, and data balancing methods. (5 Marks)

15. A reinforcement learning agent trained to solve a navigation problem fails to learn an optimal policy and shows erratic behavior. Analyze possible causes such as sparse rewards, improper exploration-exploitation balance, or incorrect reward function design. Describe debugging approaches and recommend optimization strategies like reward shaping, epsilon decay tuning, and policy optimization methods to improve learning performance. (5 Marks)

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	**30**	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	**20**	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	**30**	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	**10**	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	**10**	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

9. SVM Underperforms on a Nonlinear Dataset (5 Marks)

Problem

A Support Vector Machine (SVM) model performs poorly on a **nonlinear dataset** because of incorrect kernel selection, improper parameter tuning, or lack of feature scaling.

Possible Causes

1. **Incorrect Kernel Choice**
    - Using a **linear kernel** for nonlinear data cannot capture complex patterns.
2. **Improper Hyperparameter Tuning**
    - Incorrect values of **C** and **gamma** can lead to underfitting or overfitting.
3. **Feature Scaling Issues**
    - SVM is sensitive to the scale of features. Unscaled features may reduce accuracy.

Debugging Steps

- Visualize the dataset to check if it is nonlinear.
- Compare different kernels (Linear, Polynomial, RBF).
- Normalize or standardize the input features.
- Evaluate performance using cross-validation.
- Analyze the confusion matrix and accuracy score.

Optimization Techniques

1. **Kernel Selection**
    - Use the **RBF kernel** or **Polynomial kernel** for nonlinear datasets.
2. **Hyperparameter Tuning**
    - Tune **C** and **gamma** using **Grid Search** or **Random Search**.
3. **Feature Scaling**
    - Apply **StandardScaler** or **MinMaxScaler** before training.

Conclusion

Choosing the correct kernel, properly tuning hyperparameters, and scaling features significantly improve SVM performance on nonlinear datasets.

10. NLP Model Using RNN Suffers from Poor Long-Term Dependency Learning (5 Marks)

Problem

An RNN-based NLP model cannot effectively learn long-term dependencies in text sequences.

Causes

1. **Vanishing Gradient Problem**
    - Gradients become very small during backpropagation, preventing learning from earlier time steps.
2. **Limited Memory Capacity**
    - Standard RNNs struggle to remember information from long sequences.
3. **Long Sequential Dependencies**
    - Important information from earlier words is lost over time.

Debugging Steps

- Monitor training and validation loss.
- Check gradient values during training.
- Test the model on sequences of different lengths.
- Compare the performance of RNN with LSTM or GRU.

Solutions

1. **LSTM (Long Short-Term Memory)**
    - Uses memory cells and gates to retain long-term information.
2. **GRU (Gated Recurrent Unit)**

    - Simpler than LSTM with fewer parameters while preserving long-term dependencies.
3. **Attention Mechanism**
    - Allows the model to focus on important words instead of relying only on the hidden state.
4. **Improved Training Strategies**
    - Gradient clipping
    - Proper weight initialization
    - Learning rate scheduling
    - Dropout for regularization

Conclusion

Replacing standard RNNs with **LSTM/GRU**, adding **attention mechanisms**, and using better training strategies improve long-term dependency learning and overall NLP performance.

11. SVM Produces Low Accuracy on Training and Test Data (5 Marks)

Problem

An SVM classifier achieves **low accuracy on both training and testing datasets**, indicating that the model is underfitting or incorrectly configured.

Possible Causes

1. **Improper Kernel Selection**
    - A linear kernel may not capture nonlinear decision boundaries.
2. **Incorrect Hyperparameter Tuning**
    - Low **C** can cause underfitting.
    - Incorrect **gamma** can prevent the model from learning useful patterns.
3. **Poor Feature Scaling**
    - Features with different scales negatively affect SVM performance.
4. **Poor Quality Features**
    - Irrelevant or noisy features reduce classification accuracy.

Systematic Debugging Steps

- Inspect the dataset for missing values and outliers.
- Scale features using **StandardScaler** or **MinMaxScaler**.
- Try different kernels (Linear, Polynomial, RBF).
- Tune **C** and **gamma** using cross-validation.
- Evaluate the confusion matrix, precision, recall, F1-score, and accuracy.

Optimization Techniques

1. **Kernel Tuning**
    - Test Linear, Polynomial, and RBF kernels to find the best fit.
2. **Normalization**
    - Normalize or standardize all input features before training.
3. **Grid Search**
    - Use **GridSearchCV** to identify the optimal values of **C** and **gamma**.
4. **Cross-Validation**
    - Validate the model using k-fold cross-validation to ensure good generalization.

Conclusion

Low accuracy in both training and testing is commonly caused by incorrect kernel selection, poor hyperparameter tuning, or improper feature scaling. Applying normalization, kernel tuning, and Grid Search significantly improves SVM classification performance.